ELEG 3143 Assignment # 11

1. A random number generator produces the following dwquences of random numbers: 0.276, 0.123, 0.072, 0.324, 0.815, 0.312, 0.432, 0.283, 0.717.

   (a) Find the sample mean.

   (b) If we known the random numbers are uniformly distributed between 0.000 and 0.999, find the variance of the sample mean.

   (c) How large should the sample size be in order to obtain a sample mean whose standard deviation is no greater than 0.01?

2. (Matlab question) Generate 100 sets of 10 each of a uniformly distributed random variable extending over the interval (0, 10).

   (a) For each set of samples compute the sample mean, which is an estimate of the population mean. Thus you will get 30 different sample means.

   (b) Compute the mean of the sample means, and compare this to its theoretical value.

   (c) Compute the variance of the 100 sample means, and compare this to its theoretical value.

3. In a class of 50 students, the result of a particular exam is a true mean of 70 and a true variance of 12. It is desired to estimate the mean by sampling, without replacement, a subset of the scores.

   (a) Find the standard deviation of the sample mean if only 10 scores are used.

   (b) How large should the sample size be for the standard deviation of the sample mean to be one percentage point (out of 100)?

(c) How large should the sample size be for the standard deviation of the sample mean to be 1% of the true mean?

4. A random number generator produces the following dwquences of random numbers: 0.276, 0.123, 0.072, 0.324, 0.815, 0.312, 0.432, 0.283, 0.717. Find the sample variance if an unbiased estimator is used.

5. A Gaussian random time function with mean 2 and variance 4 is sampled so as to obtain independent sample values. How many sample values are required to obtain an unbiased estimate of the mean of the time function, with a standard deviation that is 2% of the true mean?